ADELEKE UNIVERSITY
FACULTY OF BUSINESS AND SOCIAL SCIENCES
DEPARTMENT OF ECONOMICS
B.Sc. ECONOMICS
2022/2023 FIRST SEMESTER TEST (2)

COURSE CODE: ECN 301 **COURSE TITLE: Intermediate Microeconomics**
TIME ALLOWED: 40 Minutes

INSTRUCTION: ANSWER ALL QUESTIONS

1. Why does a price-discriminating monopolist has two revenue functions?
2. Clearly explain any two reasons why firms in the perfectly competitive market are price takers
3. 4. If a household consumes three commodities A, B and C, and its income is given as I, using the tool of mathematics, state the household optimization problem.
4. An isoquant is analogous to _____ in the theory of consumer behaviour.
5. Using geometric tool of analysis where necessary, explain the rationale behind any two (2) of the following features of an indifference curve:

i. An indifference cannot touch any of the axes
ii. An indifference curve slopes downward
iii. The farther away an indifference curve is from the origin, the higher the level of satisfaction it yields.
iv. Two difference curves do not intersect.

6. In production analysis, total cost (TC) is equal to total variable cost (TVC) in the long-run. Explain.
7. Suggest two possible means by which a monopolist's activities can be controlled by the government.
8. Enumerate three possible causal factors of increasing return to scale (IRS)
9.. _____ is the locus of producer equilibrium as he moves from one level of isoquant to another in order to expand his production.
10. Concord Golden is a perfectly competitive firm whose inverse demand function is given as $P(Q) = 190 - 3q$ and its total cost function $TC = 225 + 4.5q\ 2$.

Obtain the firm's equilibrium quantity, price and profit.

10. Mr. Marvin consumption of pizza and visit to the games house are represented as X and Y respectively and her utility function is given as $U = 8XY$ subject to the constraint imposed by her income (M). If the constraint function is given as $P_xX+P_yY = M$, calculate the combination of commodities X and Y as well as λ that gives Mr. Marvin optimal satisfaction.

ii. If the prices of pizza and visit to games house are \$20 and \$60 respectively and her income is \$840, calculate the equilibrium values of X, Y and λ.

B.Sc. ECONOMICS
FIRST SEMESTER, 2022/2023 EXAMINATION
ECN 301 – INTERMEDIATE MICROECONOMICS

Lecturer: Dr. Olufunmilayo O. Jemiluyi | **Credit Unit: 3 Units**
Instruction: Attempt any four (4) questions | **Time Allowed: 2 Hours**

QUESTION ONE

(a) Enumerate any three causal factors of decreasing return to scale (DRS) in production process. **(3 marks)**

(b) Silicon Vale Plc is a firm which operates in a pure monopoly market and its product's price and cost functions are given respectively as $P = 300 - 16q$ and $C = 200 + 7.5q^2$.

i. Calculate the output and price levels that maximise the firm's profit. **(7 marks)**

ii. From the analysis in (i) above, ascertain if the sufficient condition for profit maximisation is satisfied. **(2 marks)**

iii. Calculate the optimal profit level. **(3 marks)**

(Total: 15 marks)

QUESTION TWO

Provide brief but concise answers to the following questions.

i. Explain the relevance of price elasticity of demand of products to the success of price discriminating action of monopolists. **(2 marks)**

ii. Why are firms in perfectly competitive market regarded as price takers? **(2 marks)**

iii. How can a firm enhance productivity in the short-run? **(3 marks)**

iv. Enumerate the steps involved in the application of Langrage multiplier technique to optimisation problem. **(3 marks)**

v. Differentiate between strong and weak ordering of revealed preference theory. **(5 marks)**

(Total: 15 marks)

QUESTION THREE

(a) Outline any three conditions necessary for the success of price discrimination in a monopolistic market. **(3 marks)**

(b) The utility function of a group of friends who regularly hang out at Dominos Eatery (X) and Genesis Cinema (Y) is represented as $U = 7.5XY$ and their consumption of X and Y is subject to their constrained budget line.

i. If the income constraint function is given as $P_xX + P_yY = I$, calculate the combination of commodities X and Y and the value of λ that give the friends optimal satisfaction. **(7 ½ marks)**

ii. If the unit price of hanging out at the eatery and cinema are ₦25 and ₦10, respectively, and their contributed outing sum is ₦5000, calculate the equilibrium values of X, Y and λ. **(4 ½ marks)**

(Total: 15 marks)

QUESTION FOUR

(a) With the aid of graph, explain the concept of production expansion path (PEP). **(4 ½ marks)**

(b) With the aid of geometric illustrations where necessary, explain the rationale behind any three (3) of the following features of isoquant curves:

i. an isoquant curve cannot touch any of the axes **(3 ½ marks)**

ii. an isoquant curve slopes downwardly **(3 ½ marks)**

iii. the farther away an isoquant curve is from the origin, the higher the level of output it yields. **(3 ½ marks)**

iv. two isoquant curves do not intersect. **(3 ½ marks)**

(Total: 15 marks)

QUESTION FIVE

(a) State the two basic ways by which a firm can achieve its overall economic goal. **(2 marks)**

(b) Due to the difference in price elasticity of demand for its output, a monopoly firm is able to practise price discrimination policy. If the firm faces the following demand functions; $P_1 = 80 - 2.5_{q_1}$ and $P_2 = 135 - 15_{q_2}$ in the two separate markets and its cost function is fixed as $C = 75 + 30(q_1 + q_2)$. calculate

i. The sales distribution that maximizes its profit **(5 marks)**

ii. The price level in each market **(3 marks)**

iii. The total level of profit realised by the firm **(5 marks)**

(Total: 15 marks)

QUESTION SIX

(a) Explain the economising concept and why it is relevant in economic analysis. **(2 ½ marks)**

(b) Describe the household as a basic microeconomic institution, clearly stating its economic roles and overall economising objective analysis. **(5 marks)**

(c) A winter jacket manufacturing company developed a profit model that depend on the number of jackets sold (x) and the number of hours of advertising (y), according to the function $z = f(x, y) = 48x + 96y - x^2 - 2xy - 9y^2$ subject to its budgetary constraint, $20x + 4y = 216$. Find the values of x and y that minimise the cost function. **(7 ½ marks)**

(Total: 15 marks)

FIRST SEMESTER, 2023/2024 EXAMINATI
ECN 331 - PUBLIC SECTOR ECONOMICS

Lecturer: Prof Adama, I. J. & Dr. Olopade, B. C. **Credit Units: 3**
Instruction: Attempt any four (4) questions **Time Allowed: 2 Ho**

QUESTION ONE

(a) What are public goods? (5 marks)
(b) Mention and briefly explain five (5) types of public goods. (5 marks)
(c) Briefly explain the challenges identified in public goods. (5 marks)
(Total: 15 marks)

QUESTION TWO

(a) What is public sector economics? (4 marks)
(b) Outline the history of public sector economics. (6 marks)
(c) Distinguish between private sector and public sector. (5 marks)
(Total: 15 marks)

QUESTION THREE

(a) Justify the relevance of public sector economics and policy. (5 marks)
(b) Mention and briefly explain five (5) reasons for the split up of acco nt of the budge. (5 marks)
(c) List and briefly discuss five (5) main classifications of government udgets. (5 marks)
(Total: 15 marks)

QUESTION FOUR

(a) What is public expenditure? (3 marks)
(b) Mention and briefly discuss four (4) ranges of activities that public xpenditure encompasse. (6 marks)
(c) Outline and briefly explain four (4) main reasons why government e igages in public expenditure. (6 marks)
(Total: 15 marks)

QUESTION FIVE

(a) Explain the Theory of Free Rider. (3 marks)
(b) Mention and briefly explain four (4) problems posed by Free Rider i a various context. (6 marks)
(c) Outline and briefly discuss four (4) main strategies for dealing with c allenges posed by Free Rider beha (6 marks)
(Total: 15 marks)

QUESTION SIX

Write notes on the following:
(a) Principles of government expenditure (3 marks)
(b) Wagner's Law of increasing state activities (3 marks)
(c) Wiseman Peacock Hypothesis (3 marks)
(d) Pure and impure public goods (3 marks)
(e) Causes of public expenditures (3 marks)
(Total: 15 marks)

B.Sc. ECONOMICS
FIRST SEMESTER, 2023/2024 EXAMINATION
ECN 311 – THEORY OF GROWTH AND DEVELOPMENT I

Lecturer: Dr. Olopade, Bosede Comfort | **Credit Units: 3**

Instruction: Answer any four (4) questions | **Time Allowed: 2 Hours**

QUESTION ONE

(a) What is economic growth? **(5 marks)**

(b) Briefly examine the key roles that investment plays in the process of economic growth. **(5 marks)**

(c) State five (5) assumptions of Harrod-Domar Model. **(5 marks)**

(Total: 15 marks)

QUESTION TWO

(a) Explain the Solow Growth Model. **(5 marks)**

(b) Highlight three (3) criticisms of the model. **(3 marks)**

(c) What are the implications of the model? **(2 marks)**

(d) Enumerate five (5) assumptions of the model. **(5 marks)**

(Total: 15 marks)

QUESTION THREE

(a) The obstacles to development are the factors that act as constraints to development in Less Developed Countries (LDCs). Briefly explain five (5) economic factors and (3) non-economic factors that constitute these obstacles. **(8 marks)**

(b) Briefly discuss seven (7) characteristics of LDCs. **(7 marks)**

(Total: 15 marks)

QUESTION FOUR

(a) If real Gross Domestic Product (GDP) in Nigeria for 2022 and 2023 were ₦12.7 million and ₦13.1 million, respectively, calculate the GDP growth rate in the economy. **(5 marks)**

(b) Differentiate between nominal GDP and real GDP. **(5 marks)**

(c) What do you understand by "capital formation"? **(5 marks)**

(Total: 15 marks)

QUESTION FIVE

(a) Explain the term "sustainable development". **(5 marks)**

(b) Highlight five (5) objectives of sustainable development. **(5 marks)**

(c) Enumerate five (5) policies of sustainable development. **(5 marks)**

(Total: 15 marks)

QUESTION SIX

(a) Define Todaro's core values of development. **(4 marks)**

(b) List and explain three (3) objectives of development in any society. **(6 marks)**

(c) Differentiate between economic growth and economic development. **(5 marks)**

(Total: 15 marks)

DEPARTMENT OF ECONOMICS
B.Sc. ECONOMICS
FIRST SEMESTER, 2023/2024 EXAMINATION
ECN 361 – OPERATIONS RESEARCH

Lecturer: Dr. Biyi, Oyeyemi O. **Credit Units: 3**

Instruction: Attempt any four (4) questions **Time Allowed: $2\frac{1}{2}$ Hours**

QUESTION ONE

(a) Explain the assumptions of Linear programming **(6 marks)**

(b) Solve the linear programming problem below using the graphical method:

$$\begin{aligned} \max \quad & z = 3x_1 + 2x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 100 \\ & x_1 + x_2 \leq 80 \\ & x_1 \leq 40 \\ & x_1, x_2 > 0 \end{aligned}$$

(9 marks)
(Total: 15 marks)

QUESTION TWO

Use both North West Corner method and Least Cost method to solve the transportation problem below:

Sources	Lagos	Akure	Awka	Supply
Sokoto	90	85	70	600
Kano	175	110	95	1400
Maiduguri	205	190	130	1000
Demand	1600	1050	350	

(15 marks)
(Total: 15 marks)

QUESTION THREE

Solve the decision problem below completely.

Alternative Products	Market Demand		
	High (₦)	Moderate (₦)	Low (₦)
Body Cream	1500	650	-150
Hair Cream	1200	470	-120
Hand Lotion	800	450	-95

(15 marks)
(Total: 15 marks)

QUESTION FOUR

(a) Explain the various classifications of models. **(10 marks)**

(b) What are the objectives of decision making? **(5 marks)**

(Total: 15 marks)

ADELEKE UNIVERSITY
Department of Economics
Operation Research (ECO 361) Mid Semester Test
Instruction: Attempt any two questions Time: 1 hour

1. a. Explain the assumptions of Linear programming

b. Solve the linear programming problem below using graphical method

$$\max z = 3x1 + 2x2$$

s.t. $2x1 + x2 \leq 100$

$x1 + x2 \leq 80$

$x1 \leq 40$

$x1, x2 > 0$

2. Use both North West Corner Method and Least Cost Method to solve the transportation problem below

Sources	Lagos	Akure	Awka	Supply
Sokoto	90	85	70	600
Kano	175	110	95	1400
Maiduguri	205	190	130	1000
Demand	1600	1050	350	

3 Solve the decision problem below completely.

Alternative Products	Market Demand		
	High (N)	Moderate (N)	Low (N)
Body Cream	1500	650	-150
Hair Cream	1200	470	-120
Hand Lotion	800	450	-95

ECN 321 – APPLIED STATISTICS I

Lecturer: Dr. Biyi, Oyeyemi O. **Credit Units: 3**

Instruction: Attempt any four (4) questions **Time Allowed: $2\frac{1}{2}$ Hours**

QUESTION ONE

Prepare three months moving average forecast using Table 1 below.

Table 1

Month	Jan.	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec.
Sales	400	420	425	440	390	470	510	525	560	570	600	610

(15 marks)

(Total: 15 marks)

QUESTION TWO

The total score in a professional basketball game is the sum of the scores of the two teams. An expert commentator claims that the average total score for NBA games is 202.5. A fan suspects that this is an overstatement and that the actual average total score is less than 202.5. He selects a random sample of 85 games and obtains a mean total score of 199.2 with standard deviation of 19.63. Determine, at the 5% level of significance, whether there is sufficient evidence in the sample to reject the expert commentator's claim. **(15 marks)**

(Total: 15 marks)

QUESTION THREE

It is hoped that a newly developed pain reliever will more quickly produce perceptible reduction in pain to patients after minor surgeries than a standard pain reliever. The standard pain reliever is known to bring relief in an average of 3.5 minutes with standard deviation of 2.1 minutes. To test whether the new pain reliever works more quickly than the standard one, 50 patients with minor surgeries were given the new pain reliever and their times to relief were recorded. The experiment yielded sample mean $\bar{x} = 3.1$ minutes and sample standard deviation s=1.5 minutes. Is there sufficient evidence in the sample to indicate, at the 5% level of significance, that the newly developed pain reliever does deliver perceptible relief more quickly? (Note that z=-1.645). **(15 marks)**

(Total: 15 marks)

QUESTION FOUR

(a) Four hundred randomly selected working adults in a certain State, including those who worked at home, were asked the distance from their home to their workplace. The average distance was 8.84 miles with standard deviation of 2.70 miles. Construct a 99% confidence interval for the mean distance from home to work for all residents of the State. **(7 marks)**

(b) What are the components of time series? **(8 marks)**

(Total: 15 marks)

QUESTION FIVE

Use Table 2 below to calculate the weighted average using the following indices:

(a) Laspeyres index number **(3 marks)**

(b) Paasche index number **(3 marks)**

(c) Fisher's ideal index number **(3 marks)**

(d) Marshall Edge-worth index number **(6 marks)**

(Total: 15 marks)

Table 2

Commodity	Quantities		Prices	
	1986	1990	1986	1990
A	30	70	75	360
B	40	100	160	300
C	50	150	250	960
D	15	33	180	291

QUESTION SIX

(a) Explain the probability sample techniques that you know. **(10 marks)**

(b) Differentiate with examples between open ended and close ended questionnaires. **(5 marks)**

(Total: 15 marks)

... hours

- ...ON ONE -

(a) Nigeria's external debt management strategies have varied from time to time since independence. Explain five (5) steps developed to reduce the burden of external debt in Nigeria.

(b) Briefly discuss the four (4) ...

ADELEKE UNIVERSITY, EDE

FACULTY OF BUSINESS AND SOCIAL SCIENCES

DEPARTMENT OF ECONOMICS

B.Sc. ECONOMICS

2023/2024 MID SEMESTER TEST

ECN 301 – INTERMEDIATE MICROECONOMICS

Lecturer: Mr. Eseyin, Oluwasegun **Credit Units:** 3

Time Allowed: 1 Hour

INSTRUCTION: ANSWER ALL QUESTIONS

1. (a) Discuss the meaning of production function, homogeneity of production function and return to scale

   (b) Determine the level of homogeneity and return to scale of the following production functions:

   (i) $Q = 4\sqrt{KL}$; (ii) $Q = X^2 + 6XY + 7Y^2$; (iii) $Q = X^3 - XY^4 + 3Y^3 + X^2Y^3$; (iv) $Q = \frac{3X^2}{5Y}$

2. A monopolist is faced with a linear demand curve

   $Q = 25 - \frac{1}{4}P$ and a total cost $C = 50 + 20Q$.

   (a) Determine the maximum profit of the monopolist taking into account both first and second order conditions.

   (b) Determine and interpret the elasticity of demand of the monopolist's product.

3. Use both the graphical and mathematical illustrations to describe the equilibrium of a consumer whose total money income is ₦180 and is faced with the utility function $U = 16X_1^{1/4}X_2^{3/4}$ assuming that the prices of commodities X_1 and X_2 are ₦3 and ₦5.

...allocation commission/committee experiences in Nigeria. **(9 marks)**

(b) Expand on the three (3) bases on which tax is identified. **(6 marks)**

(Total: 15 marks)

B.Sc. ECONOMICS
FIRST SEMESTER, 2023/2024 EXAMINATION
ECN 341 – NIGERIAN PUBLIC FINANCE

Lecturer: Dr. Bank-Ola, Rebecca | **Credit Units: 3**
Instruction: Attempt any four (4) questions | **Time Allowed: 2 Hours**

QUESTION ONE -

(a) Nigeria's external debt management strategies have varied from time to time since independence. Explain five (5) steps developed to reduce the burden of external debt in Nigeria. **(10 marks)**

(b) Briefly discuss the four (4) basic dimensions of public budget. **(5 marks)**

(Total: 15 marks)

QUESTION TWO -

(a) Explain three (3) principles of revenue allocation in Nigeria **(9 marks)**

(b) Describe four (4) legal instruments for the realisation of tax policy objectives in Nigeria. **(6 marks)**

(Total: 15 marks)

QUESTION THREE

(a) In a tabular form, express four (4) guidelines for dealing with public finances, emphasising on the action required by public officials for each principle. **(10 marks)**

(b) Discuss two (2) models of a State. **(5 marks)**

(Total: 15 marks)

QUESTION FOUR -

(a) Discuss five (5) potential mineral sources of revenue in Nigeria, stating one use of each and two (2) locations where each can be found. **(10 marks)**

(b) Explain the sources of federal or state government revenue in Nigeria with appropriate examples. **(5 marks)**

(Total: 15 marks)

QUESTION FIVE

(a) Describe three (3) revenue allocation commission/committee experiences in Nigeria. **(9 marks)**

(b) Expand on the three (3) bases on which tax is identified. **(6 marks)**

(Total: 15 marks)

QUESTION SIX

(a) Discuss any four (4) debt management problems in Nigeria. **(10 marks)**

(b) Explain performance budget and state three (3) of its advantages. **(5 marks)**

(Total: 15 marks